

S2

1. Course Administration & Assessment

- **Assessment Structure:** The course consists of three Multiple Choice Quizzes (MCQs), Group Case Studies, and a Final Group Project¹.
- **Case Studies:** Groups of 4 will submit a short paper and deliver a 10-minute presentation. The goal is to inform colleagues on a specific topic².
- **Final Project:** Follows a similar format to the case studies (paper + presentation) but is more extensive, taking place in the final session³³³³.
- **Group Registration:** Students must email the professor to be set up on Moodle/Zoom for submissions⁴.

2. Introduction to Machine Learning (ML)

Definition & History

- **Definition:** ML is a subfield of Artificial Intelligence (AI) focused on building systems that learn from data to identify patterns and make decisions with minimal human intervention, rather than using hard-coded rules⁵.
- **Evolution:**
 - *1950s-60s:* Symbolic AI/Expert Systems used hard-coded logic (if \$A\$ and \$B\$, then \$C\$)⁶.
 - *Modern Era:* Shifted towards probabilistic models (IBM Watson, Deep Blue, Neural Networks)⁷.

Core Terminology

- **Features (Independent Variables/Predictors):** The input data used to make predictions (e.g., temperature, time of day)⁸.
- **Labels (Dependent Variables/Outcome):** The output the model is trying to predict (e.g., ice cream sales)⁹.

- **Weights/Coefficients:** Values representing the importance or strength of a specific feature in predicting the label¹⁰¹⁰¹⁰¹⁰.

Research Hypothesis Types

- **Deductive Hypothesis (A Priori):** You start with a theory ("I believe \$X\$ affects \$Y\$") and collect data to test it. This is standard in traditional statistics and supervised learning¹¹¹¹¹¹¹¹.
- **Inductive Hypothesis (Exploratory):** You do not have a pre-existing theory. You feed data into the model to see which features are most important. This is common in unsupervised learning¹².

3. The Data Science Pipeline

A standard machine learning workflow parallels statistical research design¹³¹³¹³¹³.

1. **Data Collection (ETL):** Extract, Transform, and Load data from sources¹⁴.
2. **Cleaning & Feature Selection:** Handling missing values and selecting relevant variables. *Critical Note:* If two variables are highly correlated (multicollinearity), such as "Season" and "Temperature," the model cannot distinguish which is driving the outcome. One must be removed¹⁵¹⁵¹⁵¹⁵.
3. **Model Selection:** Choosing the algorithm based on the data type (e.g., Linear Regression for continuous data, Logistic for categorical)¹⁶.
4. **Training:** The model iterates to minimize error (variance) and "learns" the optimal weights for the features¹⁷.
5. **Evaluation:** Testing the model against a hold-out dataset (usually an 80/20 split: 80% for training, 20% for testing) to ensure it works on new, unseen data¹⁸.

4. Statistical Foundations & Algorithms

Linear Regression (Continuous Output)

- **Concept:** Fitting a line of best fit ($y = mx + c$) through data points to minimize the squared vertical distance (variance) between the points and the line¹⁹¹⁹¹⁹¹⁹.
 - y : Dependent variable (Label).
 - x : Independent variable (Feature).
 - m : Gradient/Coefficient (Strength of relationship).
 - c : Intercept (Baseline value when $x=0$).
- **OLS (Ordinary Least Squares):** The standard estimator used to calculate the line with the minimum error²⁰.

Logistic Regression (Categorical Output)

- **Concept:** Used for binary classification (0 or 1, Yes or No).
- **Sigmoid Function:** Wraps the linear equation to squash the output between 0 and 1, representing a probability.
 - Example: Predicting if "John" likes a song based on Tempo (BPM). The model outputs a probability; if >0.5 , the prediction is "Likes" (1)²¹²¹²¹²¹.

Other Algorithms

- **K-Nearest Neighbors (KNN):** Classifies a data point based on the majority vote of its nearest neighbors in the data space²².
- **Decision Trees/Random Forests:** Rules-based branching models²³.

5. Types of Machine Learning

1. **Supervised Learning:**
 - **Data:** Labeled (Input + Known Output).
 - **Goal:** Train the model to map inputs to the correct output.
 - *Example:* Teaching a machine to identify currency by giving it weight (feature) and telling it the value (label)²⁴.
2. **Unsupervised Learning:**
 - **Data:** Unlabeled.
 - **Goal:** Find hidden patterns or clusters.
 - *Example:* Clustering cricket players into "bowlers" and "batsmen" based on stats without being told their positions²⁵²⁵²⁵²⁵.
3. **Reinforcement Learning:**

- **Concept:** Reward-based learning. The system acts, receives feedback (positive or negative), and adjusts.
- *Example:* Identifying an image; if wrong, negative feedback is given, and the model adjusts²⁶.

6. Advanced Topics: NLP & Large Language Models (LLMs)

Natural Language Processing (NLP)

- **Evolution:** Moved from dictionary lookups (1950s) to probabilistic models (1990s) to Transformers (Modern)²⁷²⁷²⁷²⁷.
- **Tokenization:** Breaking text into smaller units (tokens) and assigning them numerical IDs, as models cannot understand raw text²⁸.

Word Embeddings (Vectors)

- Words are converted into **Vectors** (lists of numbers) and placed in a multidimensional space²⁹.
- **Semantic Proximity:** Words with similar meanings (e.g., "Tiger" and "Zebra") are mathematically close to each other in this space. This is calculated using **Cosine Similarity** (the angle between vectors)³⁰.

The Transformer Architecture

- **Self-Attention:** The mechanism that allows the model to look at an entire sentence at once and understand context (e.g., distinguishing "Bank" of a river vs. "Bank" for money)³¹.
- **Attention Heads:** Different "heads" focus on different aspects of language (grammar, semantic relationship, position) simultaneously. It functions mathematically like running many logistic regressions in parallel³²³²³²³².
- **Next-Token Prediction:** LLMs are essentially high-speed prediction engines. They calculate the probability of the *next* word in a sequence using a **Softmax** function (normalizing values to probabilities summing to 1)³³³³³³³³.

- **Temperature:** A parameter that controls creativity. High temperature (e.g., 1.5) = flatter distribution, more random/creative. Low temperature (e.g., 0.7) = more deterministic/focused³⁴.

7. Deep Neural Networks (DNNs)

- **Structure:** Designed to mimic the human brain.
 - **Input Layer:** Receives raw data (e.g., GDP, Health stats).
 - **Hidden Layers:** Intermediate layers where weights and biases are applied to transform inputs into abstract features.
 - **Output Layer:** The final prediction (e.g., Life Expectancy)³⁵.
- **Operation:** It is essentially a series of matrix multiplications. Inputs become outputs for one layer, which become inputs for the next³⁶³⁶³⁶³⁶.

8. Practical Application: Azure Machine Learning Demo

The course included a demonstration of **Microsoft Azure Machine Learning Studio** using the "Designer" (drag-and-drop) interface.

The Azure Workflow:

1. **Compute Resource:** Must allocate a computer in the cloud to process the job³⁷.
2. **Dataset:** Import data (e.g., "Superhero" dataset containing attributes like gender, height, eye color)³⁸.
3. **Select Columns:** Choose relevant features to analyze³⁹.
4. **Clean Missing Data:** Handle gaps (e.g., removing rows with "-99" values)⁴⁰.
5. **Split Data:** Divide into Training (80%) and Scoring/Testing (20%) sets⁴¹.
6. **Train Model:** Apply an algorithm (e.g., Linear Regression)⁴².
7. **Score & Evaluate:** Run the model and check metrics like **Coefficient of Determination (R^2)**.

- *Demo Result:* The Superhero model (predicting weight based on height/gender) had an R^2 of 0, indicating no predictive power and a need for better feature selection or data quality⁴³.

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